Name:_____

Q1 Worksheet 2

Let's talk about math.

(1) Fill in the blank. If it's raining then _____. Write your ideas below.

(2) If _____ then it is wet outside.

(3) If you see an animal in the water then it must be a fish. Is this correct? Explain.

(4) Trechette is wearing a blue shirt and gold shoes. What color is Trechette's shirt? Nashwah is wearing a purple scarf or yellow pants. What color is Nashwah's scarf?

(5) If you hear a sound under your bed then there is a monster under there. Can you conclude there a monster under your bed? From that statement, what would need to happen for you to worry about a monster under your bed?

(6) What does it mean for a statement to be true? What does it mean for a statement to be false? Is there another option?

ready Set go!

(1) What is a set? How do you describe a set? Write an example of a set in three different ways: a sentence, brackets, and a diagram.

(2) What is the union of the sets: $\{\heartsuit, \Psi, \oint, ☺\} \cup \{\Psi, ♫, \triangle, ☹, ☾, ☺\}$?

(3) What is the intersection of the sets: $\{\heartsuit, \Psi, \oint, ☺\} \cap \{\Psi, ♫, \triangle, ☹, ☾, ☺\}$? What is the intersection of the even numbers with the odd numbers?

(4) Describe the complement of the set of all cats. Is the set of all cats in the complement of the set of all cats?

(5) Consider the set that contains two students: {Harmony, Trinity}. How is that set diferent from the set $\{\text{Harmony}, \{\text{Trinity}\}\}$?

(6) What is the union of the sets: $\{\text{Trinity}\} \cup \{\text{Trinity}, \{\text{Trinity}\}\}$?

(7) What is the cardinality of the set: $\{\text{Trinity}, \{\text{Trinity}\}, \{\text{Trinity}, \{\text{Trinity}\}\}\}$